

AI Product Engineer Challenge

Build an AI-Powered Deviation Intake Module

GOAL • Build a working Deviation Management workflow for an API pharmaceutical manufacturer.

1. What You Need to Build

Build this workflow: Deviation document / text → AI extraction → Log Deviation form → AI impact & severity assistance → user review → save.

Example scenario: An API manufacturing batch has a process parameter outside its approved range. The user should be able to report the event and use AI to structure the information.

2. Reference UI

Use the screenshot below as the UI reference. You may improve the design, but keep the workflow simple and professional.

The screenshot displays the AIVOA AI Deviation Intake Module interface. The top navigation bar includes the AIVOA logo, a menu with 'QMS', 'Dashboard', 'Deviations' (active), 'CAPAs', 'Change Control', 'Audits', 'Documents', and 'Reports', and a user profile for 'Vasudha Pharma Chem Limited'.

Log Deviation Section:

- Header:** 'Log Deviation' with a 'Draft' status button. Subtext: 'Record any unexpected event, out-of-specification result or non-conformance.'
- 1. DEVIATION INFORMATION:**
 - Site / Plant:** Dropdown menu (selected: 'API Manufacturing Unit').
 - Date of Occurrence:** Text input (placeholder: 'dd-mm-yyyy').
 - Title / Short Description:** Text input (placeholder: 'e.g. OOS result for Assay in Batch ABC-001').
 - Source:** Dropdown menu (selected: 'Select source').
 - Related Product / Material:** Search input (placeholder: 'Search product or material...').
 - Batch/Lot Number:** Text input (placeholder: 'Enter batch / lot no.').
- 2. DEVIATION DETAILS:**
 - Detailed Description:** Text area (placeholder: 'Describe what happened, where, when and how it was detected...'). Character count: '0/2000'.
 - Initial Impact:** Dropdown menu (selected: 'Select impact').
 - Initial Severity:** Dropdown menu (selected: 'Select severity').
- Buttons:** 'Reset Form' and 'Save Deviation'.

AI Deviation Assistant Section:

- Header:** 'AI Deviation Assistant' with a 'BETA' label.
- Document Upload:** 'Drag & drop supporting document here or [click to browse](#)'.
- OR**
- Paste Input:** 'Paste deviation details / notes'.
- Supported Formats:** 'Supported formats: PDF, DOCX, TXT, XLS, JPG, PNG. Max file size: 10MB'.
- EXTRACTION PROGRESS:** Progress bar at 20%. Text: 'Analyzing document content and extracting key details... Please wait, this may take a few moments.'
- AI ASSISTANT:**
 - Text: 'Upload a deviation report, lab result, or paste text above. I will automatically extract the relevant details and populate the form for you.'
 - Input: 'Ask me anything about deviations...'.
 - Footer: 'AI responses may contain errors. Please verify information.'

IMPORTANT • The demo video is for the Customer Complaint module. **DO NOT build the Complaint module.** Use the video only to understand the AI workflow and interaction pattern, then build the equivalent workflow for Deviations.

3. Reference Demo Video

https://drive.google.com/file/d/1av2lzDPx8YMSzTrIz7w51HTRWBz3_5Nj/view?usp=sharing

 AIVOA- complaints module demo.mp4

4. AI Requirements

- Accept a deviation document or pasted text/email.
- Use AI to extract relevant information and populate the form.
- Provide an initial impact/severity recommendation with a short reason.
- Let the user review and edit AI-generated information before saving.

Note: You should not fill the left form manually instead utilize AI panel to fill out the left form as demonstrated in the Reference demo video

5. Technology

Frontend	Backend	AI	Database
React + Redux	Python + FastAPI	LangGraph + Groq	PostgreSQL / MySQL

Groq: gemma2-9b-it or llama-3.3-70b-versatile or any other reliable model.

6. Demo & Submission

Demo video (5–8 min): Show the working workflow first, then briefly explain the key implementation.

Show: input → AI processing → extracted form → impact/severity → review → save.

Important:

Do not upload the project, source code, demo video, assignment files, screenshots, sample documents, or any other assignment materials to GitHub or any public repository.

Submit your completed work **only through the submission form below.**

Deliverables

- **5–10 minutes demo video** explaining:(2 video submission)
 - **Working demonstration** of all implemented AI tools(pdf extraction tool, log interaction tool and edit interaction) and frontend features.
 - **Demonstrate and explain the code** by walking through the complete end-to-end workflow, starting from the user's **input** (prompt or PDF/email upload) in the frontend, showing the relevant frontend code, API endpoints, backend processing, AI/LangGraph workflow, and finally how the response populates the **Log Customer Complaint** form and **AI Copilot Risk Assessment**.

Submission Form: <https://forms.gle/TsYkUgoDdDuP1UCC9>

Deadline: 36 hours

7. What We Look For

Take ownership. Research what you need, make sensible product decisions, and build a complete working flow without waiting for every detail to be specified.

AI coding tools are allowed and encouraged. You should understand what you build and be able to explain or modify it during the interview.